

Project Planning Phase

The **Project Planning Phase** is an important stage in the development of **EduGenie – Google Gemini Powered Learning Assistant**. In this phase, the project activities, resources, schedule, team responsibilities, and development process are planned. Proper planning helps complete the project in an organized and timely manner.

1. Project Objectives

The main objectives of the project are:

- To develop an AI-based learning assistant.
- To help students solve their academic doubts.
- To provide simple and understandable explanations.
- To use Google Gemini for AI-powered responses.
- To create a simple and user-friendly application.

2. Project Activities

The major activities planned for the project include:

1. Identifying the problem and project requirements.
2. Collecting and analyzing requirements.
3. Designing the system and user interface.
4. Selecting suitable technologies and tools.
5. Developing the application.
6. Integrating the Google Gemini API.
7. Testing the system.
8. Fixing errors and improving performance.
9. Preparing documentation.
10. Final implementation and presentation.

3. Technology Planning

The following technologies can be considered for developing EduGenie:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Python or another suitable programming language
- **AI:** Google Gemini API
- **Database:** MySQL/Firebase, if required
- **Development Tools:** Visual Studio Code and a web browser

4. Time Planning

The project can be divided into different stages according to the available time:

Phase	Activity
Phase 1	Problem Identificati on
Phase 2	Requirem ent Analysis
Phase 3	Project Design
Phase 4	Developm ent
Phase 5	AI Integratio n
Phase 6	Testing
Phase 7	Document ation
Phase 8	Final Presentati on

5. Resource Planning

The resources required include:

- Computer or laptop
- Internet connection
- Development software

- Google Gemini API access
- Programming tools
- Database system, if required
- Project documentation materials

6. Risk Planning

Some possible risks include API connectivity problems, incorrect AI responses, software errors, lack of internet connectivity, and development delays. These risks can be reduced through regular testing, proper error handling, backup plans, and continuous monitoring.

7. Expected Outcome

The Project Planning Phase provides a clear roadmap for developing **EduGenie**. It helps organize the project activities, manage resources, reduce risks, and complete the application systematically. Proper planning ensures that the final system meets its intended educational purpose.